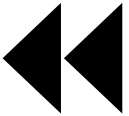


Step 1

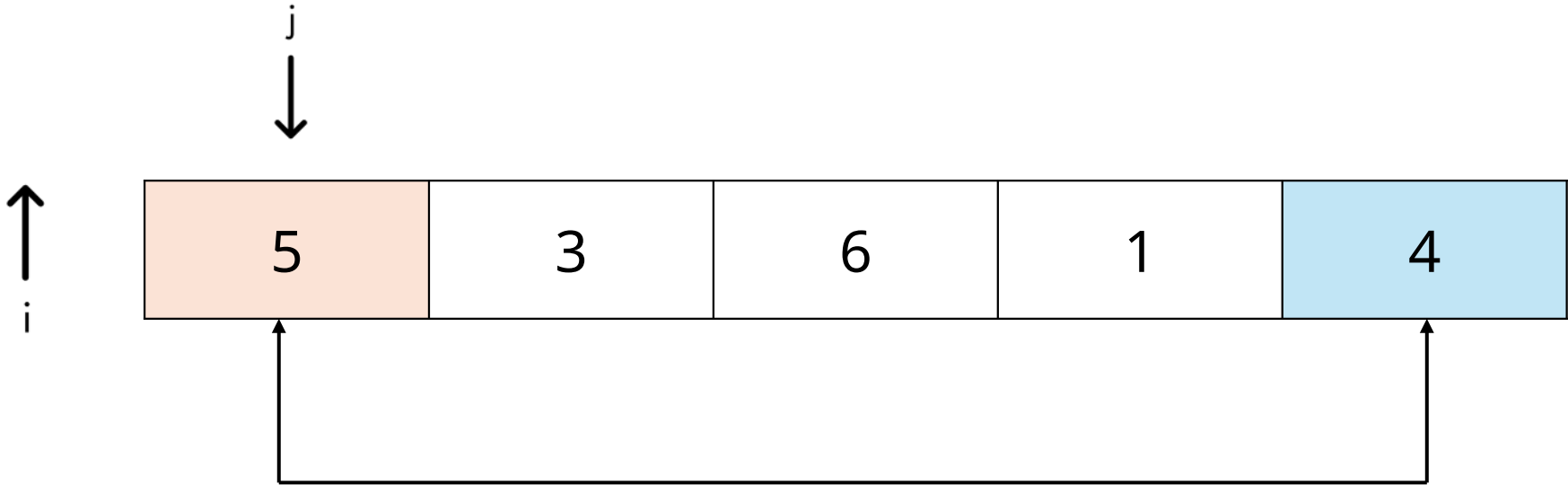
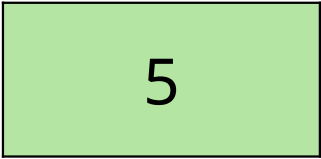


Initial pointers: $i = -1, j = 0, low = 0, high = 4$

Since, $arr[0] == pivot$

Swap $arr[j]$ with $arr[high]$

Pivot



Step 1



Pointers: $i = -1$, $j = 0$, $low = 0$, $high = 4$

Since $arr[0] < pivot$

Increase i , $i = 0$

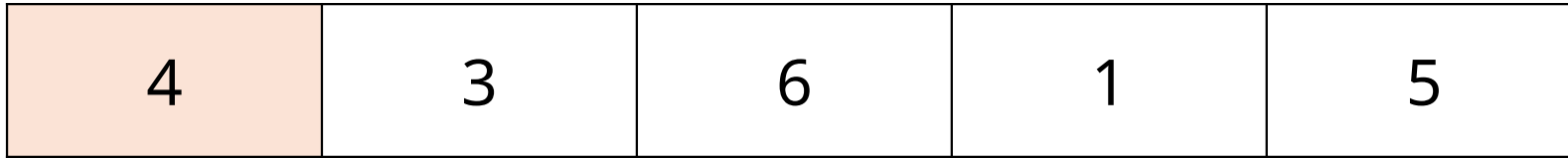
Since $i == j$, Skip swapping

Increase j

Pivot



j



i

Step 2



Pointers: $i = 0, j = 1, \text{low} = 0, \text{high} = 4$

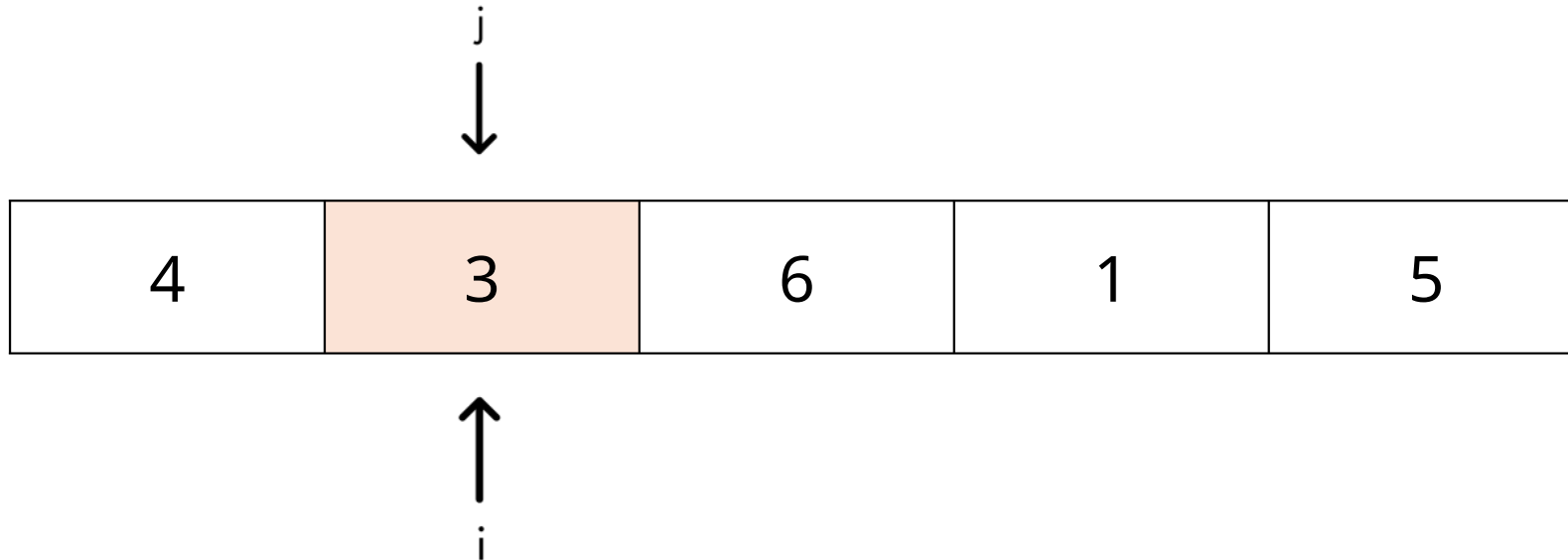
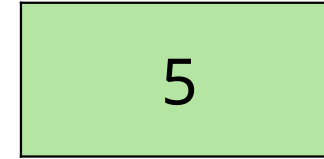
Since $\text{arr}[1] < \text{pivot}$

Increase $i, i = 1$

Since $i == j$, Skip swapping

Increase j

Pivot



Step 3



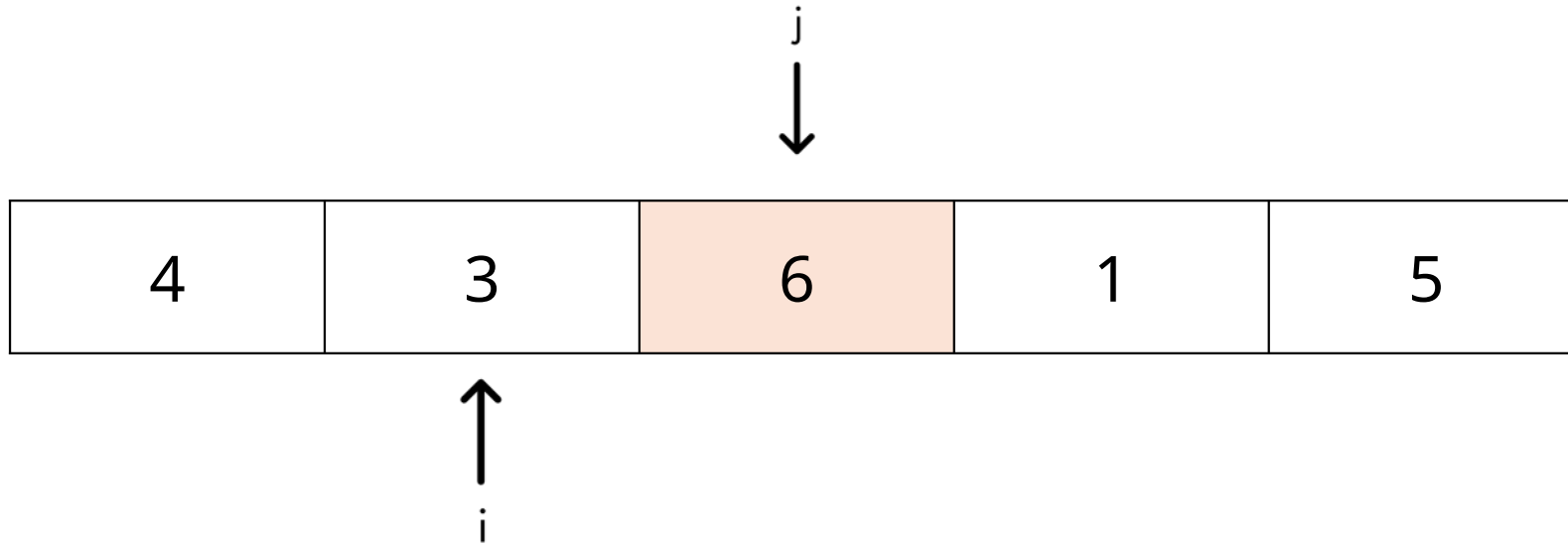
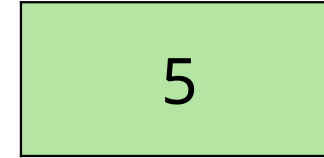
Pointers: $i = 1, j = 2, \text{low} = 0, \text{high} = 4$

Since $\text{arr}[2] > \text{pivot}$

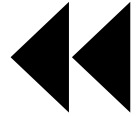
Skip swapping

Increase j

Pivot



Step 4



Pointers: $i = 1, j = 3, \text{low} = 0, \text{high} = 4$

Since $\text{arr}[3] < \text{pivot}$

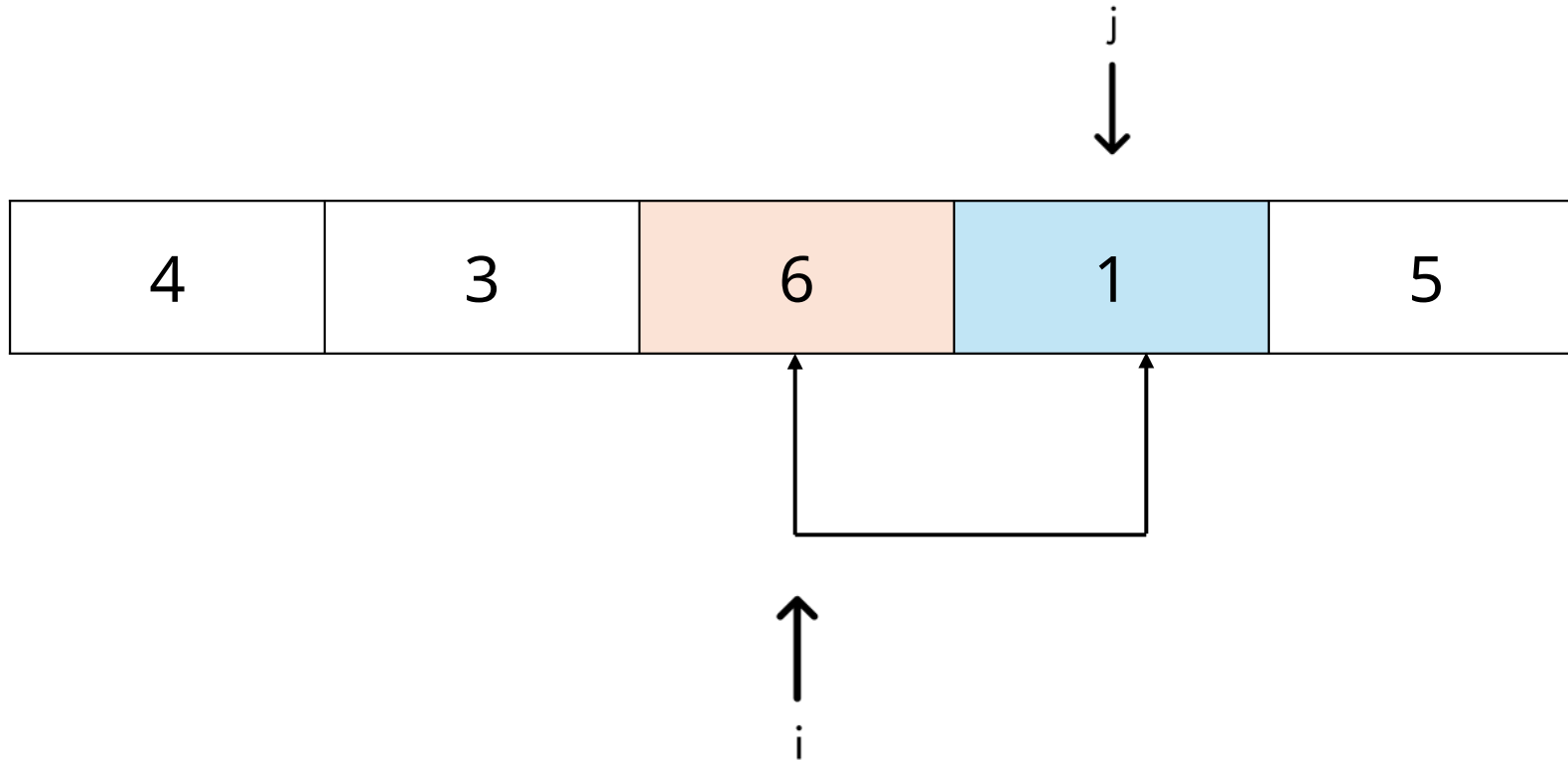
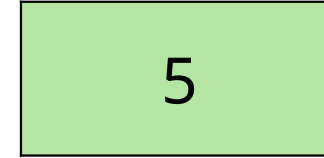
Increase $i, i = 2$

Since $i \neq j$,

Swap $\text{arr}[2]$ and $\text{arr}[3]$

Increase j

Pivot



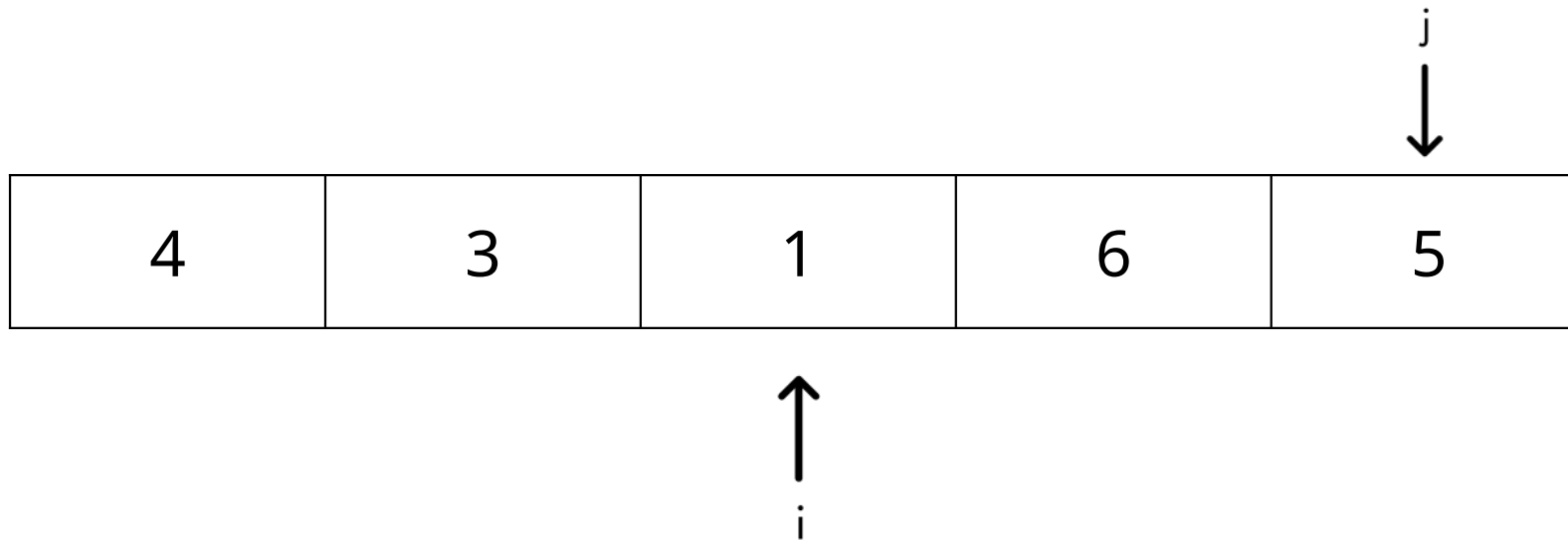
Step 5



Pointers: $i = 2$, $j = 4$, $low = 0$, $high = 4$

Since $j < high$ is false,
Exit loop

Pivot

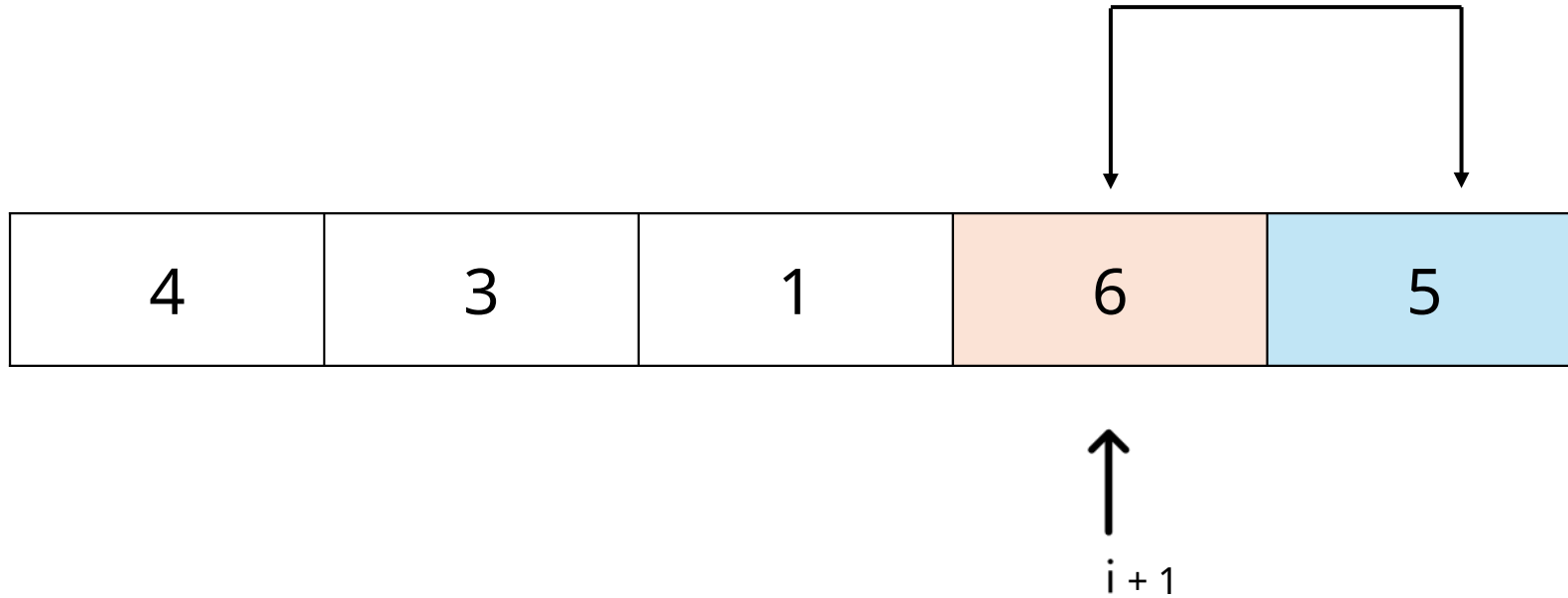
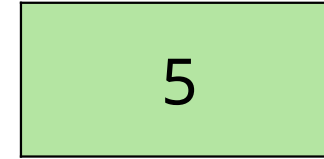


Step 5

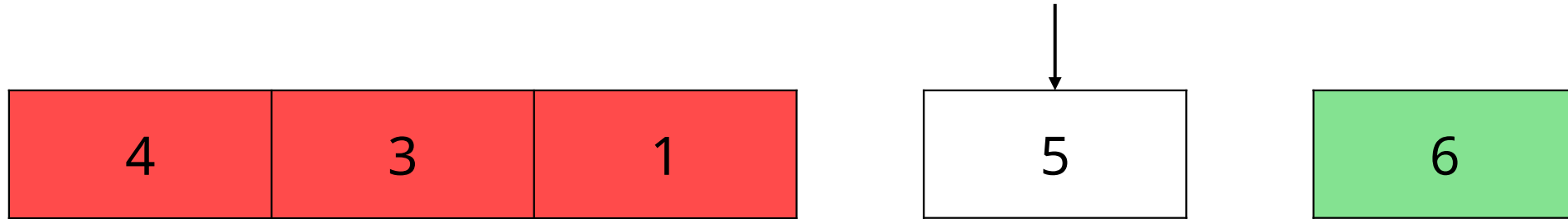


Pointers: $i = 2, j = 4, \text{low} = 0, \text{high} = 4$
Since, $i + 1 \neq \text{high}$,
Swap $\text{arr}[i + 1]$ and $\text{arr}[\text{high}]$

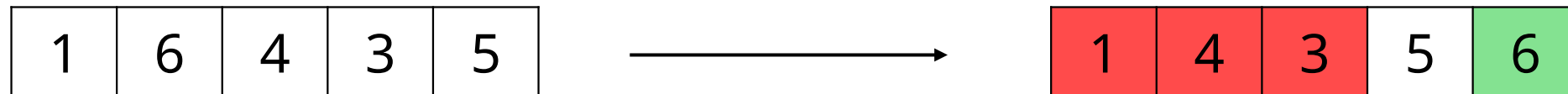
Pivot



Continue ◀◀ Now array is divided to two parts



- 1 - Return index of the pivot, which is 3
- 2 - Get array[3] as pivot, which is 5 again.
- 3 - Divide reflection array with pivot

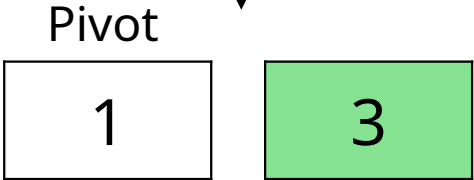


Do same steps with calling function again

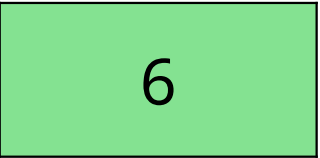
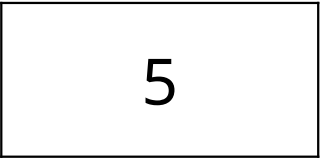
For left side use range: [low, pivotIndex - 1]

For right side use range: [pivotIndex + 1, high]

Recursion◀◀ Steps of recursion



Low >= High
Return



Low >= High
Return